Technical Report: Modeling, Identification, and Implementation of Context-Aware Battery Thermal Management: Mode-Switching MPC with SVM-HMM Driving-Context Forecasting

Esra Hicret Uzer [1] and Tomáš Haniš[2]

***Abstract*— This technical report documents the theoretical derivations, state-space formulations, parameter identification procedures, and high-fidelity verification for an advanced indirect battery thermal management system (BTMS). The multi-domain plant couples a two-phase moving-boundary vapor compression cycle, an intermediate secondary coolant circuit, and a battery cold plate. Conservation equations across dynamic boundary interfaces are derived and formatted into control-oriented state-space matrices. Parameter calibration and physical verification are established via real-time co-simulation against a high-fidelity digital twin developed in GT-Suite using vehicle test data. This document serves as the mathematical and implementation companion to the primary conference paper.**

I. INTRODUCTION

The transition from internal combustion engines to electric vehicle (EV) platforms has fundamentally transformed automotive heating, ventilation, and air conditioning (HVAC) architectures. Unlike conventional systems, modern EV thermal management must concurrently regulate cabin comfort, power electronics, and high-voltage traction batteries [1]. This multi-objective mandate creates highly dynamic cooling demands and requires scalable cooling capacities that must constantly adapt across several distinct operating points, such as ultra-fast DC charging and high-speed driving profiles. Consequently, these highly integrated, multi-mode architectures necessitate advanced control-oriented modeling approaches that can seamlessly handle varying thermal loads across the vehicle network.

To design model-based control algorithms for these systems, zero-dimensional (0D) lumped-parameter LTI thermal models are frequently employed due to their mathematical simplicity. However, lumped-parameter thermal models are inherently insufficient for accurately capturing the complex two-phase flow phenomena occurring within vapor compression cycles (VCC). Specifically, lumped approaches fail to efficiently resolve the spatial distribution of latent heat transfer during phase transitions in evaporators and condensers. To capture these critical non-linearities, moving-boundary modeling (MBM) provides a computationally tractable yet physically rigorous alternative. By dynamically tracking the spatial interfaces between subcooled liquid, two-phase, and superheated vapor regions, MBMs accurately represent the latent heat exchanges and transient dynamics that dictate system efficiency under variable capacities [2], [3].

Furthermore, for advanced control strategies like Model Predictive Control (MPC) to be safely deployed, the underlying state-space plant model must be verified against high-fidelity representations [4]. Establishing a verified digital twin addresses the critical gap between theoretical mathematical formulations and practical embedded software implementation. Developing a comprehensive digital twin in GT-Suite provides a meticulously calibrated, physics-based environment capable of simulating complex multi-domain interactions, including multi-phase refrigerants and conjugate heat transfer. Utilizing GT-Suite for software-in-the-loop co-simulation ensures that empirical parameter identification mimics the actual vehicle dynamics during transient scenarios.

To combine high-level stochastic driving contexts and real-time optimal thermal control, this technical report documents the complete mathematical foundations of a context-aware Battery Thermal Management System (BTMS). This work provides a non-equilibrium moving-boundary formulation for the vapor compression cycle and heat dissipation path from the heat generating elements through the cold plate and coolant cycle and parameter specifications calibrated and verified against the vehicle digital twin in GT-Suite.

II. SYSTEM ARCHITECTURE

The thermal architecture is structured into three thermally coupled physical domains, illustrated in Fig. 1:

Vapor Compression Cycle: Contains an electric variable-speed compressor, condenser/gas cooler, internal heat exchangers, three expansion valves (EXVs) and eight solenoid valves, an indirect refrigerant-to-coolant chiller, and a cabin air evaporator.

Coolant Circuit: Circulates an ethylene glycol-water mixture driven by dual centrifugal pumps to manage the high-voltage battery cold plates, traction motor, and inverter.

Air-side: Manages fresh/recirculated cabin climate comfort via blower motors and positive temperature coefficient (PTC) heaters.

III. MATHEMATICAL MODELING

A. 1D Two-Phase Moving-Boundary Refrigerant Dynamics

The heat exchangers undergo transient phase changes categorized into three distinct moving zones as it is in

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[1]Esra Hicret Uzer is with the Faculty of Electrical Engineering, Czech Technical University in Prague, Prague 120 00, Czech Republic uzeresra@fel.cvut.cz

[2]Tomáš Haniš is with the Faculty of Electrical Engineering, Czech Technical University in Prague, Prague 120 00, Czech Republic tomas.hanis@cvut.cz

NOMENCLATURE

A	Frontal cross-sectional area [m^2]
a_v	Expansion valve opening fraction [$-$]
C_{com}	Compressor clearance volume ratio [$-$]
C_p, c_p	Specific isobaric heat capacity [$\mathrm{J}/(\mathrm{kg}\cdot\mathrm{K})$]
C_v, A_v	Flow coefficient [$-$] and orifice area [m^2]
$h, \bar{h}$	Local and mean specific enthalpies [J/kg]
L_{ci}, L_{ei}	Condenser/evaporator zone lengths [m]
$m, \dot{m}$	Mass [kg] and mass flow rate [kg/s]
n	Isentropic compression efficiency [$-$]
p_c, p_e	Condenser/evaporator pressures [Pa]
P_{pump}	Electrical power draw of coolant pump [W]
$\dot{Q}$	Heat transfer exchange rate [W]
$R_{contact}$	Core/plate thermal contact resistance [K/W]
T	Thermodynamic temperature [K or $^\circ$C]
UA	Overall heat transfer conductance [W/K]

Greek symbols

α	Convective heat transfer coefficient [$\mathrm{W}/(\mathrm{m}^2\cdot\mathrm{K})$]
ΔP_{sys}	Coolant hydraulic circuit pressure drop [Pa]
ε	Effectiveness of heat exchanger [$-$]
η_{is}	Isentropic compression efficiency [$-$]
η_{pump}	Electromechanical pump efficiency [$-$]
$\rho, \bar{\rho}$	Local and lumped mean fluid densities [kg/m^3]
τ_d	Hydraulic transport time delay [s]
ω_{com}	Compressor rotational angular speed [rad/s]

Subscripts

amb	Ambient environmental condition
c, ch, e	Condenser, chiller and evaporator
cp, f	Battery cold plate and secondary coolant fluid
d, s	Compressor discharge and suction ports
i, o	Inlet/internal and outlet/external conditions
inv, mot	inerter and motor
lg	Liquid-to-gas saturation transition
r, w, a	Refrigerant, tube wall, and air domains
$1, 2, 3$	Superheated, two-phase, and subcooled zones

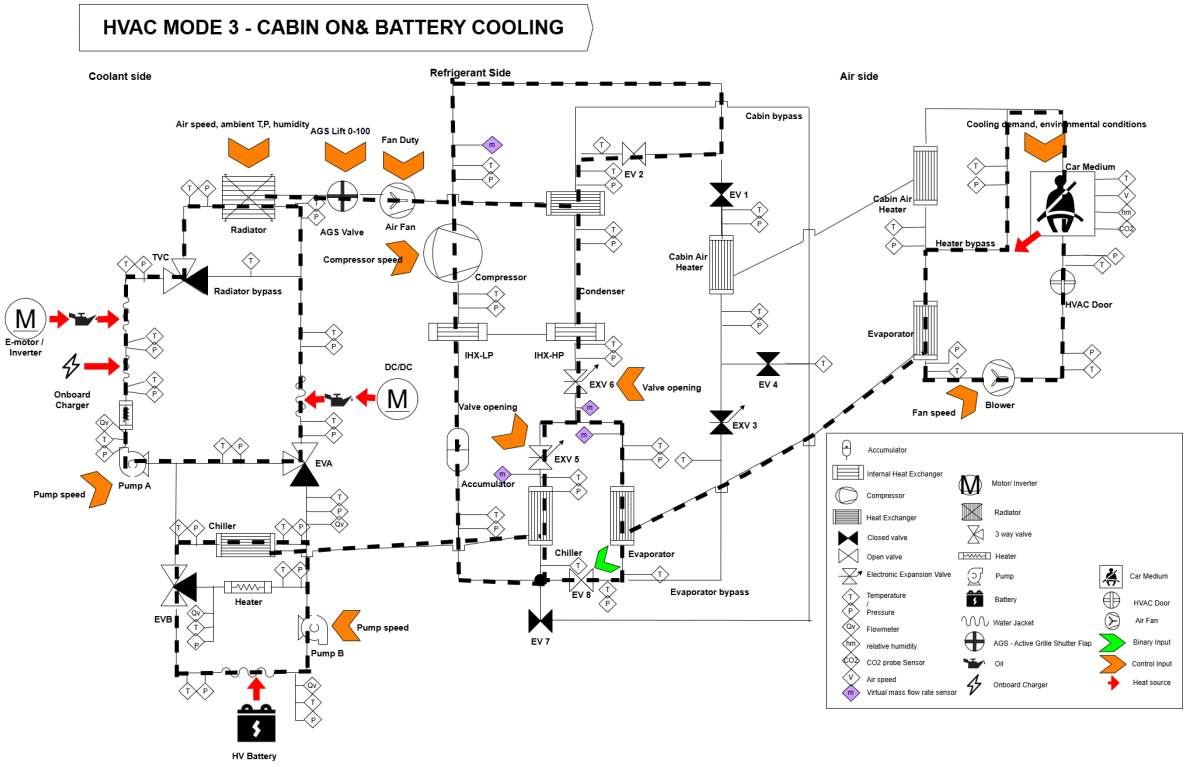


Fig. 1. Multi-loop vehicle thermal topology: intermediate secondary coolant circuit (left), two-phase refrigerant loop (center), and passenger cabin/radiator air interface (right).

Fig. 2:: superheated vapor (L_1), two-phase mixture (L_2), and subcooled liquid (L_3). Spatial integration of the continuity and energy conservation laws yields the boundary dynamics:

$$\frac{\partial \rho}{\partial t} + \frac{\partial (\rho u)}{\partial z} = 0, \tag{1}$$

$$\frac{\partial (\rho A h - A p)}{\partial t} + \frac{\partial (\dot{m} h)}{\partial z} = \pi D_i \alpha_i (T_w - T_r), \tag{2}$$

$$(C_p \rho A)_w \frac{\partial T_w}{\partial t} = \pi D_i \alpha_i (T_r - T_w) + \pi D_o \alpha_o (T_a - T_w). \tag{3}$$

Integrating across time-varying spatial boundaries $z \in [z_j(t), z_{j+1}(t)]$ via Leibniz's rule produces the nonlinear descriptor state-space realization:

$$A_k(x_k)\dot{x}_k = f_k(x_k, u_k), \quad k \in \{c, e\}. \tag{4}$$

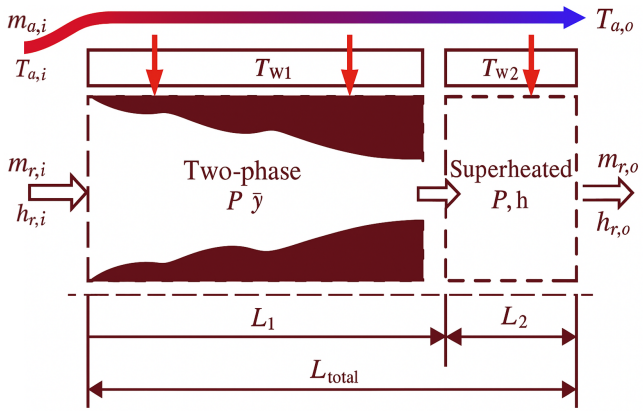


Fig. 2. Input-output thermodynamic boundary interaction for the moving-boundary heat exchanger formulation.

The condenser state vector is:

$$x_c = \begin{bmatrix} L_{c1} & p_c & L_{c2} & h_{co} & T_{cw1} & T_{cw2} & T_{cw3} \end{bmatrix}^T. \quad (5)$$

The structural matrix $A_c \in \mathbb{R}^{7\times7}$ is formatted with strict column width bounding:

$$A_c = \begin{bmatrix} a_{11} & a_{12} & 0 & 0 & 0 & 0 & 0 \\ a_{21} & a_{22} & a_{23} & a_{24} & 0 & 0 & 0 \\ a_{31} & a_{32} & a_{33} & a_{34} & 0 & 0 & 0 \\ a_{41} & a_{42} & a_{43} & a_{44} & 0 & 0 & 0 \\ a_{51} & 0 & 0 & 0 & L_{c1} & 0 & 0 \\ a_{61} & 0 & a_{63} & 0 & 0 & L_{c2} & 0 \\ a_{71} & 0 & a_{73} & 0 & 0 & 0 & L_{c3} \end{bmatrix}, \quad (6)$$

where $L_{c3} = L_c - L_{c1} - L_{c2}$. Its analytical entries are:

$$\begin{aligned}
a_{11} &= A_c\bar{\rho}_1(\bar{h}_1 - h_1), \\
a_{12} &= A_cL_{c1}\left(\frac{\partial(\bar{\rho}_1\bar{h}_1)}{\partial p_c} - \frac{\partial\bar{\rho}_1}{\partial p_c}h_1 - 1\right), \\
a_{21} &= A_c\bar{\rho}_3(h_2 - \bar{h}_3), \\
a_{22} &= A_cL_{c3}\left(\frac{\partial(\bar{\rho}_3\bar{h}_3)}{\partial p_c} - \frac{\partial\bar{\rho}_3}{\partial p_c}h_2 - 1\right), \\
a_{23} &= A_c\bar{\rho}_3(h_2 - \bar{h}_3), \\
a_{24} &= \frac{1}{2}A_cL_{c3}\left(\bar{\rho}_3 + \frac{\partial\rho_3}{\partial h}\bigg|_{p_c}(\bar{h}_3 - h_2)\right), \\
a_{31} &= A_c(\bar{\rho}_1 - \bar{\rho}_3), \\
a_{32} &= A_c\left(L_{c1}\frac{\partial\bar{\rho}_1}{\partial p_c} + L_{c2}\frac{\partial\rho_{lg}}{\partial p_c} + L_{c3}\frac{\partial\bar{\rho}_3}{\partial p_c}\right), \\
a_{33} &= A_c(\rho_{lg} - \bar{\rho}_3), \\
a_{34} &= \frac{1}{2}A_cL_{c3}\frac{\partial\rho_3}{\partial h}\bigg|_{p_c}, \\
a_{41} &= A_c(\bar{\rho}_1h_1 - \bar{\rho}_3h_2), \\
a_{42} &= A_c\left[L_{c1}\frac{\partial\bar{\rho}_1}{\partial p_c}h_1 + L_{c2}\left(\frac{\partial(\rho h)_{lg}}{\partial p_c} - 1\right) + L_{c3}\frac{\partial\bar{\rho}_3}{\partial p_c}h_2\right], \\
a_{43} &= A_c\left((\rho h)_{lg} - \bar{\rho}_3h_2\right), \\
a_{44} &= \frac{1}{2}A_cL_{c3}\frac{\partial\rho_3}{\partial h}\bigg|_{p_c}h_2,
\end{aligned}$$

$$\begin{aligned}
a_{51} &= -(T_{cw2} - T_{cw1}), \quad a_{61} = (T_{cw2} - T_{cw1}), \\
a_{63} &= -(T_{cw3} - T_{cw2}), \quad a_{71} = (T_{cw3} - T_{cw2}), \\
a_{73} &= (T_{cw3} - T_{cw2}).
\end{aligned}$$

The right-hand non-equilibrium forcing vector $f_c \in \mathbb{R}^{7\times1}$ is:

$$f_c = \begin{bmatrix} \dot{m}_{ci}(h_{ci} - h_1) - \dot{Q}_{cr1} \\ -\dot{Q}_{cr3} - \dot{m}_{co}(h_{co} - h_2) \\ \dot{m}_{ci} - \dot{m}_{co} \\ \dot{m}_{ci}h_1 - \dot{m}_{co}h_2 - \dot{Q}_{cr2} \\ (\dot{Q}_{cr1} - \dot{Q}_{ca1})/(c_pm/L_c)_w \\ (\dot{Q}_{cr2} - \dot{Q}_{ca2})/(c_pm/L_c)_w \\ (\dot{Q}_{cr3} - \dot{Q}_{ca3})/(c_pm/L_c)_w \end{bmatrix}. \quad (7)$$

For the refrigerant evaporator (or chiller), superheated vapor emerges at the outlet, omitting the subcooled liquid zone. The reduced state vector is $x_e = [p_e, L_{e2}, h_{eo}, T_{ew2}, T_{ew3}]^T$. The dynamic matrix $A_e \in \mathbb{R}^{5\times5}$ is:

$$A_e = \begin{bmatrix} a_{11} & a_{12} & a_{13} & 0 & 0 \\ a_{21} & a_{22} & a_{23} & 0 & 0 \\ a_{31} & a_{32} & a_{33} & 0 & 0 \\ 0 & a_{42} & 0 & L_{e2} & 0 \\ 0 & a_{52} & 0 & 0 & L_{e3} \end{bmatrix}, \quad (8)$$

with the analytical entries defined as:

$$\begin{aligned}
a_{11} &= A_eL_{e3}\left(\frac{\partial(\bar{\rho}_{e3}\bar{h}_{e3})}{\partial p_e} - \frac{\partial\bar{\rho}_{e3}}{\partial p_e}h_{e2} - 1\right), \\
a_{12} &= A_e\bar{\rho}_{e3}(h_{e2} - \bar{h}_{e3}), \\
a_{13} &= \frac{1}{2}A_eL_{e3}\left(\bar{\rho}_{e3} + \frac{\partial\rho_{e3}}{\partial h}\bigg|_{p_e}(\bar{h}_{e3} - h_{e2})\right), \\
a_{21} &= A_e\left(L_{e2}\frac{\partial\rho_{elg}}{\partial p_e} + L_{e3}\frac{\partial\bar{\rho}_{e3}}{\partial p_e}\right), \\
a_{22} &= A_e(\rho_{elg} - \bar{\rho}_{e3}), \\
a_{23} &= \frac{1}{2}A_eL_{e3}\frac{\partial\rho_{e3}}{\partial h}\bigg|_{p_e}, \\
a_{31} &= A_e\left[L_{e2}\left(\frac{\partial(\rho h)_{elg}}{\partial p_e} - 1\right) + L_{e3}\frac{\partial\bar{\rho}_{e3}}{\partial p_e}h_{e2}\right], \\
a_{32} &= A_e\left((\rho h)_{elg} - \bar{\rho}_{e3}h_{e2}\right), \\
a_{33} &= \frac{1}{2}A_eL_{e3}\frac{\partial\rho_{e3}}{\partial h}\bigg|_{p_e}h_{e2}, \\
a_{42} &= -(T_{ew3} - T_{ew2}), \quad a_{52} = (T_{ew3} - T_{ew2}).
\end{aligned}$$

The right-hand side vector $f_e \in \mathbb{R}^{5\times1}$ is:

$$f_e = \begin{bmatrix} \dot{Q}_{er3} - \dot{m}_{eo}(h_{eo} - h_{e2}) \\ \dot{m}_{ei} - \dot{m}_{eo} \\ \dot{Q}_{er2} + \dot{m}_{ei}h_{ei} - \dot{m}_{eo}h_{e2} \\ (\dot{Q}_{ea2} - \dot{Q}_{er2})/(c_pm/L_e)_{ew} \\ (\dot{Q}_{ea3} - \dot{Q}_{er3})/(c_pm/L_e)_{ew} \end{bmatrix}. \quad (9)$$

Compressor mass flow rate and enthalpy rise are calculated under adiabatic assumptions with clearance:

$$\dot{m}_{com} = \omega_{com}V\rho\left[1 + C_{com} - C_{com}\left(\frac{p_c}{p_e}\right)^{\frac{1}{n}}\right], \quad (10)$$

$$h_d = h_s + \frac{h_{is} - h_s}{\eta_{is}}. \quad (11)$$

The mass flow through the electronic expansion valve (EEV) is modeled as:

$$\dot{m}_v = a_v C_v A_v \sqrt{2\rho_{co}(p_c - p_e)}, \quad a_v \in [0,1]. \tag{12}$$

B. Coolant Dynamics

Heat transfer from the battery cell core (T_c) and casing surface (T_s) into the aluminum cold plate (T_{cp}) and fluid (T_f) is governed by:

$$\dot{T}_{cp} = \frac{1}{m_{cp}c_{p,cp}} \left[\frac{T_s - T_{cp}}{R_{contact}} - \alpha_f A_f (T_{cp} - T_{f,avg}) \right], \tag{13}$$

$$\begin{aligned} \dot{T}_{f,out} = \frac{1}{m_f c_{p,f}} & \left[\dot{m}_f c_{p,f} (T_{f,in} - T_{f,out}) \right. \\ & \left. + \alpha_f A_f (T_{cp} - T_{f,avg}) \right], \end{aligned} \tag{14}$$

where $T_{f,avg} = 0.5(T_{f,in} + T_{f,out})$.

Transport delays across inter-component piping are modeled as a first-order Padé approximation of the pure delay τ_d:

$$\dot{T}_{chiller,in} = \frac{1}{\tau_d}(T_{f,out} - T_{chiller,in}), \quad \tau_d = \frac{\rho_f V_{pipe}}{\dot{m}_f}. \tag{15}$$

The rejection of heat in the refrigerant evaporator is formulated by the Effectiveness-NTU method (ε -NTU):

$$\begin{aligned} \dot{Q}_{chiller} = & \left[1 - \exp\left(-\frac{U A_{chiller}}{\dot{m}_f c_{p,f}} \right) \right] \\ & \times \dot{m}_f c_{p,f} (T_{chiller,in} - T_{r,e}), \end{aligned} \tag{16}$$

$$T_{chiller,out} = T_{chiller,in} - \frac{\dot{Q}_{chiller}}{\dot{m}_f c_{p,f}}. \tag{17}$$

Pump hydraulic parasitic power consumption is evaluated as:

$$P_{pump} = \frac{\Delta P_{sys} \dot{m}_f}{\rho_f \eta_{pump}}. \tag{18}$$

C. Combined State Vector

$$x = \begin{bmatrix} x_c^T & x_{ch}^T & x_e^T & x_{bat}^T & x_{cl}^T & x_{cab}^T \end{bmatrix}^T \in \mathbb{R}^{n_x} \tag{19}$$

Refrigerant side:

$$x_c = \begin{bmatrix} L_{c1} & p_c & L_{c2} & h_{co} & T_{cw1} & T_{cw2} & T_{cw3} \end{bmatrix}^T, \tag{20}$$

$$x_{ch} = \begin{bmatrix} p_{ch} & L_{ch2} & h_{ch,o} & T_{chw2} & T_{chw3} \end{bmatrix}^T, \tag{21}$$

$$x_e = \begin{bmatrix} p_e & L_{e2} & h_{e,o} & T_{ew2} & T_{ew3} \end{bmatrix}^T. \tag{22}$$

Coolant side:

$$x_{bat} = \begin{bmatrix} T_c & T_s & T_{cp} & T_{f,out} \end{bmatrix}^T, \tag{23}$$

$$x_{cl} = \begin{bmatrix} T_{chiller,in} & T_{mot} & T_{inv} \end{bmatrix}^T. \tag{24}$$

Air side:

$$x_{cab} = \begin{bmatrix} T_{cab} \end{bmatrix}. \tag{25}$$

$$\underbrace{\begin{bmatrix} A_c(x_c) & & & \\ & A_e(x_{\text{ch}}) & & \\ & & A_e(x_{\text{e}}) & \\ & & & I_9 \end{bmatrix}}_{\mathcal{A}(x)} \dot{x} = \mathcal{F}(x,z,u,d) = \begin{bmatrix} f_c \\ f_{ch} \\ f_e \\ f_{bat} \\ f_{cl} \\ f_{cab} \end{bmatrix} \tag{26}$$

$$\mathbf{z} = \begin{bmatrix} \dot{m}_{\text{com}},\ h_d,\ \dot{m}_{v,1..3},\ \dot{m}_{\text{ci}},\ \dot{m}_{\text{co}},\ \dot{m}_{\text{ei}},\ \dot{m}_{\text{eo}},\ \dot{Q}_{\text{chiller}},\ T_{\text{chiller,out}},\ P_{\text{pump}},\ \dot{Q}_{\text{cr/ca/er/ea}} \end{bmatrix} \tag{27}$$

$$u = \begin{bmatrix} \omega_{\text{com}} & a_{v1} & a_{v2} & a_{v3} & \omega_{p1} & \omega_{p2} & \dot{m}_{\text{blw}} & P_{\text{PTC}} & \omega_{\text{fan}} \end{bmatrix}^T, \quad y = \begin{bmatrix} T_c & T_{\text{cab}} & T_{f,\text{out}} & SH_{\text{ch}} & SH_{\text{e}} & SC_c & p_c & p_e \end{bmatrix}^T \tag{28}$$

$$d = \begin{bmatrix} T_{\text{amb}} & \dot{Q}_{\text{bat,gen}} & \dot{Q}_{\text{mot,gen}} & \dot{Q}_{\text{inv,gen}} & \dot{Q}_{\text{solar}} & v_{\text{veh}} \end{bmatrix}^T, \quad \sigma \in \{0,1\}^8 \quad \text{(solenoid valve modes)} \tag{29}$$

IV. CALIBRATION AND MODEL PARAMETERS

The baseline parameters are calibrated against the manufacturer specifications / vehicle test results for the subjected BEV platform, summarized in Table I.

TABLE I
MODEL SPECIFICATIONS AND PARAMETERS

Parameter	Value
Cold plate	Aluminum
Coolant fluid	egl5050
Refrigerant	R744
Total electromechanical pump efficiency	0.72
Isentropic compression efficiency	0.82

V. GT-SUITE DIGITAL TWIN & CO-SIMULATION

The plant model is evaluated against a high-fidelity digital twin modeled in GT-Suite which was calibrated with the experimental vehicle data. The GT-Suite simulation calculates empirical heat transfer correlations based on local thermodynamic properties, temperature and pressure drops.

Communication between implemented controller on MATLAB/Simulink and the GT-Suite digital twin runs through compiled S-functions adhering to the Functional Mock-up Interface (FMI) co-simulation standard. The simulation timestep is set to $T_s = 0.01\,\text{s}$ with zero-order hold (ZOH) extrapolations to maintain real-time execution bounds to simulate thermal dynamics.

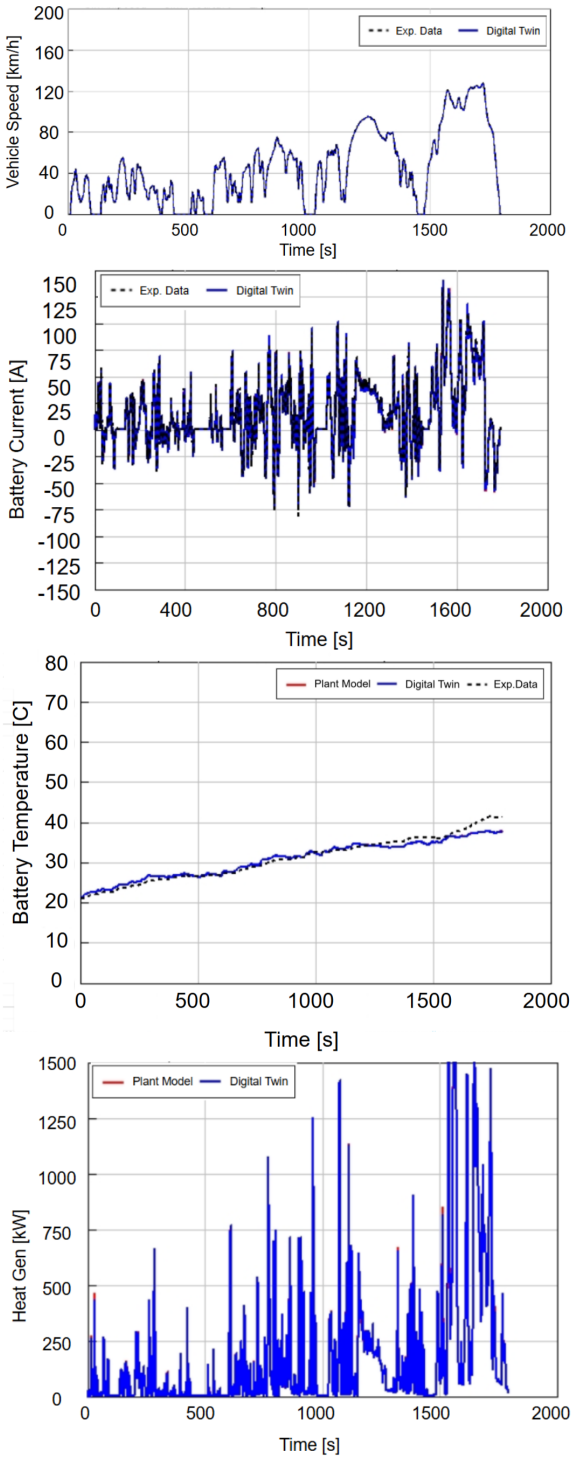


Fig. 3. Validation of the digital twin under a dynamic driving cycle. Experimental and simulated responses are compared for vehicle speed, battery current, battery temperature, and battery heat generation rate. The digital twin accurately captures the transient electrical and thermal behavior across varying operating conditions.

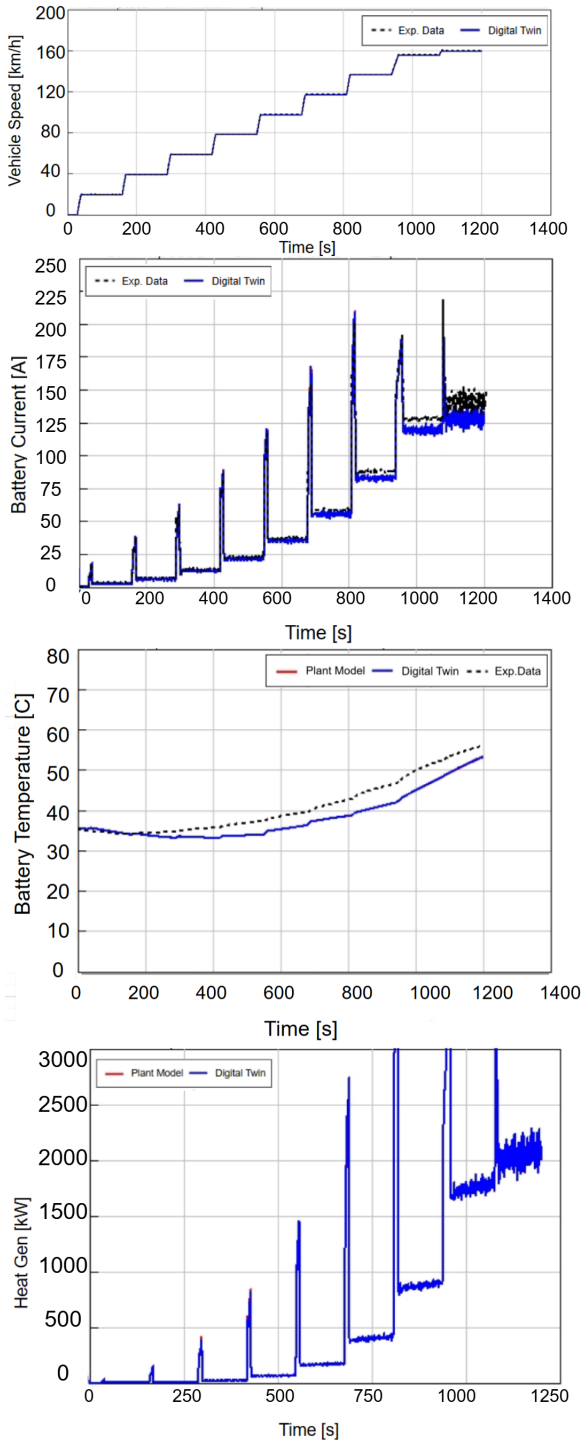


Fig. 4. Validation of the digital twin against experimental data under increasing vehicle-speed demand. The comparison includes vehicle speed, battery current, battery temperature, and heat generation, showing satisfactory reproduction of the measured thermal and electrical behavior.

VI. CONCLUSIONS

This technical companion report presents the governing thermodynamic and energy, mass conservation equations and heat dissipation mechanism through refrigerant cycle along with parameter specifications for an indirect BTMS. This vehicle model is calibrated and verified with several driving cycle cases. Future work and accompanying main conference publications integrate this foundation into context-aware predictive control architectures tested against wide driving cycles.

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REFERENCES

[1] D. Leighton, "Combined fluid loop thermal management for electric drive vehicle range improvement," *SAE Int. J. Passeng. Cars - Mech. Syst.*, vol. 8, no. 2, pp. 711–720, 2015, doi: 10.4271/2015-01-1709.

[2] B. D. Eldredge, B. P. Rasmussen, and A. G. Alleyne, "Moving-boundary heat exchanger models with variable outlet phase," *J. Dyn. Sys., Meas., Control*, vol. 130, no. 6, p. 061003, Nov. 2008, doi: 10.1115/1.2977466.

[3] B. P. Rasmussen and A. G. Alleyne, "Dynamic modeling and advanced control of air conditioning and refrigeration systems," Ph.D. dissertation, Dept. Mech. Eng., Univ. Illinois Urbana-Champaign, Urbana, IL, 2005.

[4] Y. Xie, C. Wang, X. Hu, X. Lin, Y. Zhang, and W. Li, "An MPC-based control strategy for electric vehicle battery cooling considering energy saving and battery lifespan," *IEEE Trans. Veh. Technol.*, vol. 69, no. 12, pp. 14657–14669, Dec. 2020, doi: 10.1109/TVT.2020.3033502.